

Notes on Gauge Theory

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0 Prologue: Electromagnetism and $U(1)$ Gauge Theory

0.1 Maxwell's equations

We begin the story with Maxwell's equations. Recall that Maxwell's equations in vacuum are

$$\begin{cases} \operatorname{div} \mathbf{E} = 0, \\ \operatorname{div} \mathbf{B} = 0, \\ \operatorname{curl} \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \\ \operatorname{curl} \mathbf{B} = \frac{\partial \mathbf{E}}{\partial t}. \end{cases}$$

Here

$$\begin{aligned} \mathbf{E} &= (E_x, E_y, E_z) \quad \text{is the electric field,} \\ \mathbf{B} &= (B_x, B_y, B_z) \quad \text{is the magnetic field.} \end{aligned}$$

Each component is a function of the spacetime coordinates (t, x, y, z) .

Below we will list some concepts and notations commonly used in physics:

- 4-vector potential A_μ ¹

$$A_\mu = (\phi, A_x, A_y, A_z),$$

where ϕ is the *scalar potential* and $\mathbf{A} = (A_x, A_y, A_z)$ is the *vector potential*.

- Electromagnetic field strength tensor $F_{\mu\nu}$

$$F_{\mu\nu} = \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & B_z & -B_y \\ E_y & -B_z & 0 & B_x \\ E_z & B_y & -B_x & 0 \end{pmatrix}.$$

- Relation between A_μ and $F_{\mu\nu}$

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \implies \begin{cases} \mathbf{E} = \operatorname{grad} \phi - \frac{\partial \mathbf{A}}{\partial t}, \\ \mathbf{B} = \operatorname{curl} \mathbf{A}. \end{cases}$$

- Gauge transformation of the 4-vector potential

$$A_\mu \mapsto A_\mu + \partial_\mu \chi \implies \begin{cases} \phi \mapsto \phi + \frac{\partial \chi}{\partial t}, \\ \mathbf{A} \mapsto \mathbf{A} + \operatorname{grad} \chi. \end{cases}$$

where χ is an arbitrary smooth function. One can check that this transformation leaves the electromagnetic field strength tensor $F_{\mu\nu}$ invariant.

¹ $\mu = 0, 1, 2, 3$ are the spacetime indices, where 0 corresponds to the time coordinate t , and 1, 2, 3 correspond to the spatial coordinates x, y, z .

²Here we use the notation $\partial_0 := \frac{\partial}{\partial t}$, $\partial_1 := \frac{\partial}{\partial x}$, $\partial_2 := \frac{\partial}{\partial y}$, $\partial_3 := \frac{\partial}{\partial z}$.

Although Maxwell's equations are already quite concise, we can still transform them into an even more elegant expression using the language of differential forms.

Let $\mathbb{R}^{3,1}$ be the spacetime with coordinates (t, x, y, z) , equipped with the Minkowski metric

$$ds^2 = dt^2 - dx^2 - dy^2 - dz^2.$$

For simplicity, we set the speed of light $c = 1$.

The Hodge star operator $*$ on $\Omega^*(\mathbb{R}^{3,1})$ is defined by

$$\alpha \wedge *\beta = \langle \alpha, \beta \rangle dt \wedge dx \wedge dy \wedge dz, \quad \forall \alpha, \beta \in \Omega^*(\mathbb{R}^{3,1}).$$

Explicitly, we have

$$\begin{aligned} *1 &= dt \wedge dx \wedge dy \wedge dz, \\ *dt &= dx \wedge dy \wedge dz, \\ *dx &= dt \wedge dy \wedge dz, \\ *dy &= dt \wedge dz \wedge dx, \\ *dz &= dt \wedge dx \wedge dy, \\ *(dt \wedge dx) &= -dy \wedge dz, \\ *(dt \wedge dy) &= -dz \wedge dx, \\ *(dt \wedge dz) &= -dx \wedge dy, \\ *(dx \wedge dy) &= dt \wedge dz, \\ *(dy \wedge dz) &= dt \wedge dx, \\ *(dz \wedge dx) &= dt \wedge dy, \\ *(dt \wedge dx \wedge dy) &= dz, \\ *(dt \wedge dy \wedge dz) &= dx, \\ *(dt \wedge dz \wedge dx) &= dy, \\ *(dx \wedge dy \wedge dz) &= dt, \\ *(dt \wedge dx \wedge dy \wedge dz) &= -1. \end{aligned}$$

It is direct to check that

$$*^2 = (-1)^{p+1} \quad \text{on } \Omega^p(\mathbb{R}^{3,1}).$$

Now we can introduce the electromagnetic field strength 2-form

$$F = -dt \wedge (E_x dx + E_y dy + E_z dz) + B_x dy \wedge dz + B_y dz \wedge dx + B_z dx \wedge dy.$$

Then

$$\begin{aligned} dF &= dt \wedge \left[(\partial_x E_y - \partial_y E_x) dx \wedge dy + (\partial_y E_z - \partial_z E_y) dy \wedge dz + (\partial_z E_x - \partial_x E_z) dz \wedge dx \right] \\ &\quad + dt \wedge \left(\partial_t B_x dy \wedge dz + \partial_t B_y dz \wedge dx + \partial_t B_z dx \wedge dy \right) \\ &\quad + \left(\partial_x B_x + \partial_y B_y + \partial_z B_z \right) dx \wedge dy \wedge dz, \end{aligned}$$

Thus

$$\begin{cases} \operatorname{div} \mathbf{B} = 0, \\ \operatorname{curl} \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}. \end{cases} \iff dF = 0.$$

On the other hand,

$$*F = E_x dy \wedge dz + E_y dz \wedge dx + E_z dx \wedge dy + dt \wedge (B_x dx + B_y dy + B_z dz).$$

So

$$\begin{aligned} d * F &= -dt \wedge \left[(\partial_x B_y - \partial_y B_x) dx \wedge dy + (\partial_z B_x - \partial_x B_z) dz \wedge dx + (\partial_y B_z - \partial_z B_y) dy \wedge dz \right] \\ &\quad + dt \wedge \left(\partial_t E_x dy \wedge dz + \partial_t E_y dz \wedge dx + \partial_t E_z dx \wedge dy \right) \\ &\quad + \left(\partial_x E_x + \partial_y E_y + \partial_z E_z \right) dx \wedge dy \wedge dz. \end{aligned}$$

Thus

$$\begin{cases} \operatorname{div} \mathbf{E} = 0, \\ \operatorname{curl} \mathbf{B} = \frac{\partial \mathbf{E}}{\partial t}. \end{cases} \iff d * F = 0.$$

Since $*$: $\Omega^p(\mathbb{R}^{3,1}) \rightarrow \Omega^{4-p}(\mathbb{R}^{3,1})$ is an isomorphism,

$$d * F = 0 \iff *d * F = 0.$$

Therefore,

$$\begin{cases} \operatorname{div} \mathbf{E} = 0, \\ \operatorname{div} \mathbf{B} = 0, \\ \operatorname{curl} \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \\ \operatorname{curl} \mathbf{B} = \frac{\partial \mathbf{E}}{\partial t}. \end{cases} \iff \begin{cases} dF = 0, \\ d * F = 0. \end{cases}$$

Here $d^* = *d*$ is the dual of d .

In Riemannian geometry, a p -form α is called *harmonic* if $d\alpha = 0$ and $d^*\alpha = 0$. Thus Maxwell's equations in vacuum corresponds to a harmonic 2-form F .

Poincaré lemma says all closed forms on $\mathbb{R}^{3,1}$ are exact. So if $dF = 0$, there exists a 1-form A such that

$$F = dA.$$

If we write

$$A = \phi dt + A_x dx + A_y dy + A_z dz,$$

Then

$$F = dA \implies \begin{cases} E_x = \partial_x \phi - \partial_t A_x, \\ E_y = \partial_y \phi - \partial_t A_y, \\ E_z = \partial_z \phi - \partial_t A_z, \\ B_x = \partial_y A_z - \partial_z A_y, \\ B_y = \partial_z A_x - \partial_x A_z, \\ B_z = \partial_x A_y - \partial_y A_x. \end{cases} \implies \begin{cases} \mathbf{E} = \operatorname{grad} \phi - \frac{\partial \mathbf{A}}{\partial t}, \\ \mathbf{B} = \operatorname{curl} \mathbf{A}. \end{cases}$$

It is easy to see that A is not unique. In fact, if we transform $A \mapsto A + d\chi$ for some smooth function χ , then

$$F \mapsto dA + d^2\chi = dA = F.$$

Explicitly,

$$A \mapsto A + d\chi \implies \begin{cases} \phi \mapsto \phi + \partial_t \chi, \\ A_x \mapsto A_x + \partial_x \chi, \\ A_y \mapsto A_y + \partial_y \chi, \\ A_z \mapsto A_z + \partial_z \chi. \end{cases} \implies \begin{cases} \phi \mapsto \phi + \frac{\partial \chi}{\partial t}, \\ \mathbf{A} \mapsto \mathbf{A} + \nabla \chi. \end{cases}$$

We can summarize the above discussion in a table:

Physics	Differential Geometry
4-vector potential A_μ	1-form A
Electromagnetic field strength tensor $F_{\mu\nu}$	2-form F
$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$	$F = dA$
Gauge transformation $A_\mu \mapsto A_\mu + \partial_\mu \chi$	$A \mapsto A + d\chi$
Maxwell's equations	$dF = 0, d^*F = 0$

0.2 $U(1)$ gauge theory

Let $\psi(x)$ be a complex-valued function on $\mathbb{R}^{3,1}$, which represents the wave function of a matter field. Here x denotes the spacetime coordinates. In physics, $|\psi(x)|^2$ represents the probability amplitude of finding the particle at a given point in spacetime. If we transform

$$\psi(x) \mapsto \psi'(x) = e^{i\chi} \psi(x)$$

for some real constant χ . This transformation is called a *global $U(1)$ gauge transformation*. Since the probability amplitude $|\psi'(x)|^2 = |\psi(x)|^2$ remains the same, the physics is invariant under global $U(1)$ gauge transformations.

Now we want to promote the global $U(1)$ gauge symmetry to a local one. That is, we want the physics to be invariant under the transformation

$$\psi(x) \mapsto \psi'(x) = e^{i\chi(x)} \psi(x)$$

for some smooth function $\chi \in C^\infty(\mathbb{R}^{3,1})$.

For some physical reason, we want the derivative of ψ to be covariant under transformations. This holds in the case of global $U(1)$ gauge transformations, since

$$d\psi \mapsto d\psi' = e^{i\chi} d\psi.$$

However, in the case of local $U(1)$ gauge transformations,

$$d\psi \mapsto d\psi' = e^{i\chi(x)} d\psi + (de^{i\chi(x)})\psi.$$

The extra term $(de^{i\chi(x)})\psi$ breaks the covariance of the derivative.

To fix this problem, we introduce a 1-form A and define the *covariant derivative* by

$$\nabla\psi = d\psi + iA\psi.$$

Under the local $U(1)$ gauge transformation,

$$\nabla \mapsto \nabla' = d + iA',$$

we want $\nabla\psi$ to be covariant, i.e.

$$\nabla\psi \mapsto \nabla'\psi' = e^{i\chi(x)}\nabla\psi.$$

Then by

$$\begin{aligned}\nabla'\psi' &= d\psi' + iA'\psi' = e^{i\chi(x)}d\psi + ie^{i\chi(x)}(A' + d\chi)\psi, \\ e^{i\chi(x)}\nabla\psi &= e^{i\chi(x)}(d\psi + iA\psi) = e^{i\chi(x)}d\psi + ie^{i\chi(x)}A\psi,\end{aligned}$$

we must have

$$A' = A - d\chi.$$

Ignoring the sign, this is exactly the gauge transformation of the electromagnetic potential A in electromagnetism. Thus, the electromagnetic potential A can be interpreted as a gauge field that ensures the covariance of the derivative under local $U(1)$ gauge transformations.